DP Biology – D3.2

Inheritance

- What patterns of inheritance are seen in plants and animals?
- What happens at the molecular level to produce these patterns?

SL statements D3.2.1–15. Textbook pp. 725–746.
Image: peas in their pods, Mendel's model organism. Bill Ebbesen, CC BY-SA 3.0.

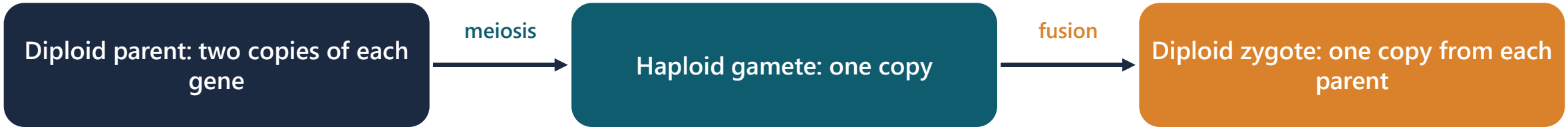


Objectives for each class

Class	Statements	By the end of the class, you can...
1	D3.2.1–5	Explain inheritance through gametes, use genotype, phenotype and dominance terms, and predict crosses with Punnett grids
2	D3.2.6–10	Explain plasticity and phenylketonuria, multiple alleles, ABO groups, incomplete dominance and codominance
3	D3.2.11–15	Explain sex determination and sex linkage, read pedigrees, and describe continuous variation with box plots

How inheritance works

D3.2.1



An **autosomal** gene is on any chromosome except a sex chromosome. Body cells have two copies; gametes have one.

So each parent passes on **one** of its two copies. This is common to all eukaryotes that reproduce sexually.

Link

Meiosis (D2.1) separates homologous chromosomes, so each gamete gets one allele of each gene.

Genetic crosses in flowering plants

D3.2.2



Gregor Mendel. Public domain.

A cross needs **pollination**: pollen (male gametes) moved to the stigma. Peas can also **self-fertilize**.

For a controlled cross, pollen from a chosen plant is moved by hand.

Term	Meaning
P	The parents
F1	Offspring of the P generation
F2	Offspring of F1 plants crossed or selfed

Why it matters

Mendel's pea crosses gave the first clear laws of inheritance; breeders still use crosses today.

Genes, alleles and genotype

D3.2.3

Term	Meaning	Example
Gene	A length of DNA that influences a characteristic	Flower colour
Allele	A version of a gene	Purple or white
Genotype	The pair of alleles inherited	Pp
Homozygous	Two identical alleles	PP or pp
Heterozygous	Two different alleles	Pp

Common mistake

Everyone has the ABO gene; people differ in their alleles of it.

Phenotype: genotype and environment

D3.2.4

The **phenotype** is what you can observe. It depends on genotype, environment, or both.

Genotype only	Environment only	Both
ABO blood group	A scar	Height
Haemophilia	A tattoo	Skin colour
Colour blindness	Language spoken	Body mass

Activity

Sort into the three groups: ability to taste bitterness, eye colour, an accent, adult height, a broken bone, type 2 diabetes. Which were hardest, and why?

Dominant and recessive alleles

D3.2.5

Genotype	Alleles	Phenotype
YY	Both make a working protein	Yellow
Yy	One working allele is enough	Yellow
yy	No working protein	Green

A **dominant** allele has the same effect in one or two copies. A **recessive** allele only shows when homozygous.

Why YY and Yy look the same

Often the dominant allele codes for a working protein and the recessive one does not. One copy makes enough.

Worked example: a Punnett grid

P: yellow (YY) × green (yy). All F1 are Yy, yellow. Two F1 plants are crossed:

Gametes	Y	y
Y	YY: yellow	Yy: yellow
y	Yy: yellow	yy: green

1. Gametes of each parent go along the sides
2. Fill in the offspring genotypes
3. Genotypes: 1 YY : 2 Yy : 1 yy
4. Phenotypes: 3 yellow : 1 green

Try it

Short fur (S) is dominant to long (s). Two heterozygous cats breed. Predict the ratio, and explain why a litter of four may not match it.

Working with data: a monohybrid cross

D3.2.5

A heterozygous yellow pea plant was self-pollinated. These counts are invented for practice; Mendel's results showed the same pattern.

Phenotype	Seeds
Yellow	612
Green	198
Total	810

1. Calculate the ratio of yellow to green.
2. Which allele is dominant? Explain with a grid.
3. Calculate the expected numbers for 3 : 1.
4. Suggest why the result is not exactly 3 : 1.

Practice: what is its genotype?

D3.2.5

Unknown purple

A purple-flowered pea could be PP or Pp.

Test cross

It is crossed with a white (pp) plant.

Results

All 40 offspring are purple.

Questions

Deduce the genotype of the purple plant. What ratio would you expect if it were Pp? Why is the recessive homozygote always used in a test cross?

Check-in: class 1

D3.2.1–5

1. Explain why gametes carry one copy of each gene.
2. Distinguish between gene and allele.
3. Define homozygous and heterozygous.
4. Explain why YY and Yy have the same phenotype.
5. Tt × tt: predict the ratio of offspring.

How to use this

Answer without your notes, then check against slides 3–10.

Phenotypic plasticity



Water flea: grows a defensive helmet near predators. Paul Hebert, CC BY 2.5.

Plasticity: developing traits that suit the environment by changing **gene expression**. The genotype is unchanged, and the change may reverse.

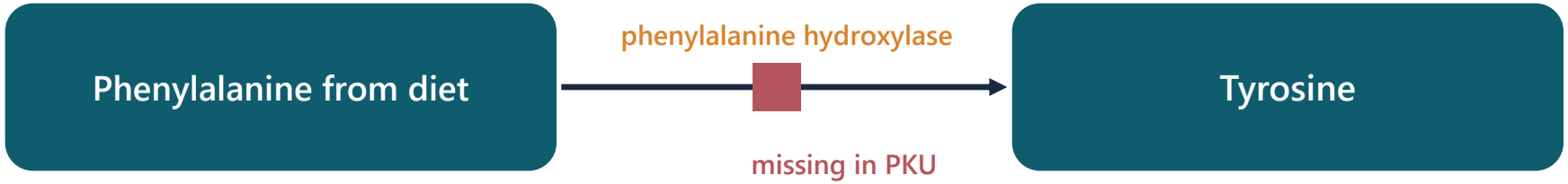
Examples: muscles grow with training; skin darkens in sun.

Common mistake

Plasticity is not mutation or natural selection. The alleles stay the same.

Phenylketonuria: a recessive disease

D3.2.7



PKU is an autosomal recessive disease: the enzyme is not made. Phenylalanine builds up and harms the developing brain. A heterozygote is a **carrier**.

Newborns are screened, and a low-phenylalanine diet prevents most harm.

Key idea

Same genotype, different outcome: the diet (environment) changes the phenotype.

SNPs and multiple alleles

D3.2.8

Allele	Part of base sequence	Change
1	A T G G C A T T C G	Original
2	A T G A C A T T C G	SNP at base 4
3	A T G G C A T T T G	SNP at base 10

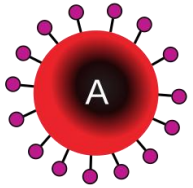
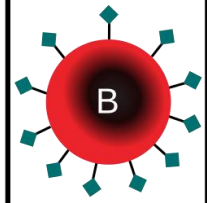
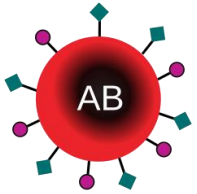
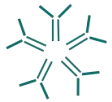
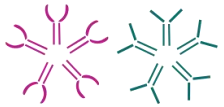
A **gene pool** is all the alleles in a population. SNPs make many alleles of one gene, but each person inherits only **two**.

Common mistake

Multiple alleles exist in a population, not in one person.

ABO blood groups

D3.2.9

	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies in plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in red blood cell	 A antigen	B antigen	 A and B antigens	None

ABO blood types. InvictaHOG, public domain.

Genotype	Group	Note
$I^A I^A$ or $I^A i$	A	I^A dominant to i
$I^B I^B$ or $I^B i$	B	I^B dominant to i
$I^A I^B$	AB	Codominant
i i	O	Recessive

Try it

A group A woman and a group B man have a group O child. Deduce all three genotypes and the other groups later children could have.

Incomplete dominance and codominance

D3.2.10



Mirabilis jalapa. Juni from Kyoto, CC BY 2.0.

	Incomplete dominance	Codominance
Heterozygote	Intermediate phenotype	Both phenotypes
Example	Red × white gives pink	$I^A I^B$ gives group AB

Worked example

Red ($C^R C^R$) × white ($C^W C^W$) gives pink F1 ($C^R C^W$). Pink × pink gives 1 red : 2 pink : 1 white, so every genotype can be told from its colour.

Practice: more than two alleles

D3.2.9–10

AB × O

Predict the blood groups of the children.

Pink × red

Mirabilis: predict the colours and ratio.

Who's the parent?

A child is group O. Could a group AB man be the father?

Show your working

Write the parents' genotypes, the gametes, and a Punnett grid for each.

Check-in: class 2

D3.2.6–10

1. Define phenotypic plasticity, with an example.
2. Explain why PKU can appear in children of unaffected parents.
3. Explain how many alleles of a gene one person can have.
4. Explain why group AB shows codominance.
5. Distinguish incomplete dominance from codominance.

How to use this

Answer without your notes, then check against slides 12–17.

Sex determination and sex chromosomes

D3.2.11



Gametes	X	Y
X	XX	XY
X	XX	XY

All eggs carry an X; half of sperm carry a Y. So the **sperm** decides the sex chromosomes of the zygote.

The X carries far more genes than the Y.

Human male karyotype. NHGRI, public domain.

Sex linkage

Males have one X, so one recessive allele on the X shows in their phenotype.

Haemophilia: a sex-linked disorder

D3.2.12

Clotting X^H, haemophilia X^h. Carrier mother × unaffected father:

Gametes	X^H	Y
X^H	$X^H X^H$: girl	X^H Y: boy
X^h	$X^H X^h$: carrier girl	X^h Y: affected boy

A clotting factor is missing, so blood clots slowly. The allele is **recessive** and on the **X**.

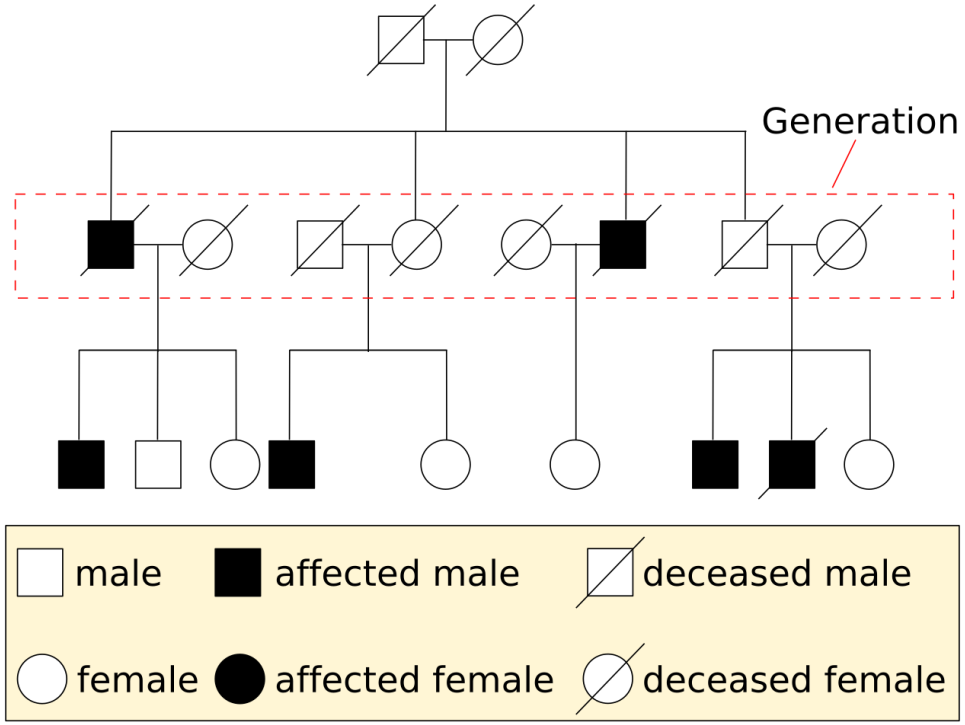
Far more males are affected. A son gets his X from his mother, never his father.

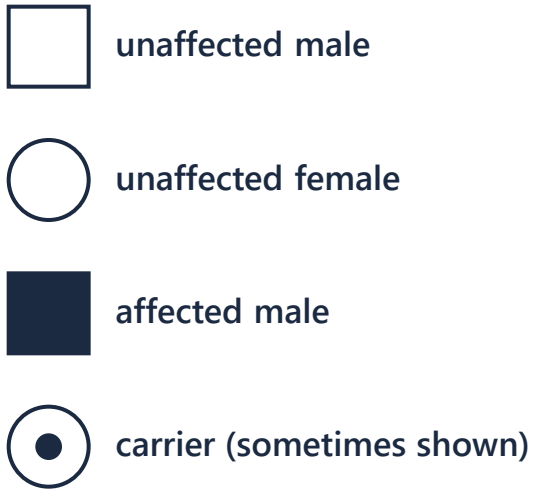
Try it

An affected man and a non-carrier woman have children. Predict the sons' and daughters' phenotypes.

Reading pedigree charts

D3.2.13





An example pedigree chart. YassineMrabet, public domain.

Lines

Horizontal lines join parents; vertical lines lead down to their children.

Deducing the pattern of inheritance

D3.2.13

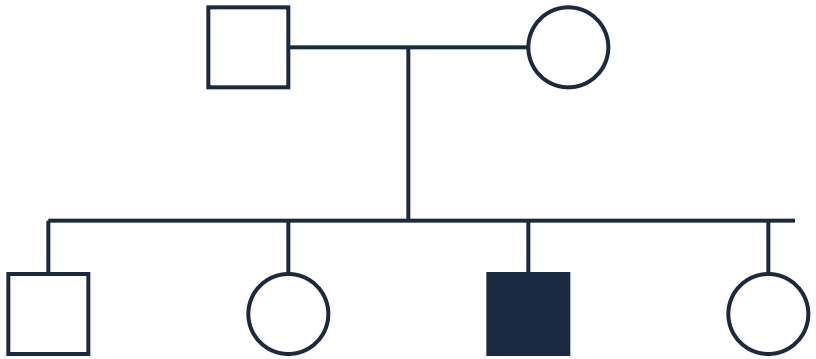
Pattern	Clues in the pedigree
Autosomal recessive	Unaffected parents can have an affected child of either sex
Autosomal dominant	Every affected person has an affected parent
X-linked recessive	Mostly males; passed from carrier mothers; never father to son

Close relatives

Relatives are more likely to carry the same recessive allele from a shared ancestor. This is the genetic basis for rules against marriage between close relatives in many societies.

Practice: read this pedigree

Generation I



Generation II

1. Is the allele dominant or recessive? Explain.
2. Give the genotypes of the parents.
3. What is the chance their next child is affected?
4. Can you rule out X-linkage from this chart alone?

Continuous variation

D3.2.14

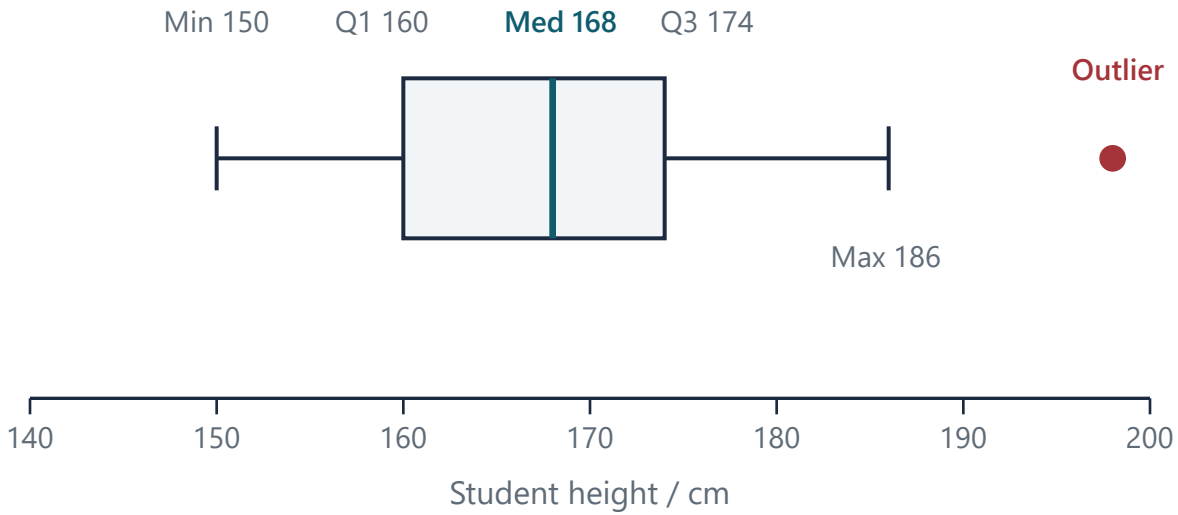
	Discrete	Continuous
Values	Separate categories	Any value in a range
Example	ABO blood group	Height, skin colour
Controlled by	Usually one gene	Many genes and environment

Polygenic: many genes each add a small effect, giving a bell-shaped curve. Describe the centre with **mean**, **median** and **mode**.

Worked example

Heights 158, 160, 160, 164, 168: mean 162, median 160, mode 160 cm.

Box-and-whisker plots



Invented values, for practice.

A box plot shows the **median**, **quartiles**, **range** and any **outliers**.

IQR = Q3 − Q1. An outlier is more than 1.5 × IQR beyond a quartile.

Worked example

IQR = 174 − 160 = 14; 1.5 × 14 = 21. Upper limit 174 + 21 = 195, so 198 cm is an outlier, and the whisker stops at 186 cm.

Working with data: comparing two groups

D3.2.14–15

Hand span of two classes, in cm. The values are invented for practice.

Class	Min	Q1	Median	Q3	Max	Outliers
Class 1	16	18	19	20	22	none
Class 2	15	17	18	21	23	28

1. Calculate the IQR of each class.
2. Check that 28 cm is an outlier.
3. Compare the two distributions.

Check-in: class 3

D3.2.11–15

1. Explain how sex is determined in humans.
2. Explain why haemophilia is more common in males.
3. State one clue that an allele is recessive in a pedigree.
4. Distinguish discrete from continuous variation.
5. Explain how to identify an outlier on a box plot.

How to use this

Answer without your notes, then check against slides 19–26.

Common mistakes to avoid

Not this	But this
Dominant alleles are more common.	Dominance says nothing about frequency.
A 3 : 1 ratio means exactly 3 in every 4.	It is a probability; real numbers vary.
Males can carry haemophilia.	Males either have it or lack the allele.
Codominance and incomplete dominance are the same.	Codominance: both show; incomplete: a blend.
Genotype alone decides phenotype.	Environment also affects phenotype.

Summary: inheritance

D3.2.1–15

	Key idea
D3.2.1–4	Haploid gametes, diploid zygote; phenotype = genotype + environment
D3.2.5	Dominant and recessive; Punnett grids, 3 : 1
D3.2.6–7	Plasticity; PKU, an autosomal recessive
D3.2.8–10	Multiple alleles, ABO, codominance, incomplete dominance
D3.2.11–12	XX / XY; haemophilia is X-linked recessive
D3.2.13	Pedigrees reveal the pattern of inheritance
D3.2.14–15	Polygenic traits; box-and-whisker plots

Check that you can...

1. **Construct** Punnett grids and **predict** ratios
2. **Distinguish** genotype from phenotype, dominant from recessive
3. **Explain** plasticity and phenylketonuria
4. **Explain** multiple alleles, codominance and incomplete dominance
5. **Explain** sex linkage using haemophilia
6. **Deduce** inheritance patterns from pedigrees
7. **Represent** continuous variation with box plots

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Two carriers of phenylketonuria have a child. State the probability that the child has the disease.
2. Distinguish between incomplete dominance and codominance.
3. Explain why a son cannot inherit haemophilia from his father.

Image credits

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Gregor Mendel 2.jpg – Unknown author – Public domain

Daphnia pulex.png – (Photo: Paul Hebert) – CC BY 2.5

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